

Unified Import Flow Planner Document

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1. Purpose

This document defines the updated planning requirements for the unified import flow that processes both YouTube Music and Spotify export data. It replaces the legacy Spotify import logic, introduces artwork caching, and adds real-time progress tracking during import.

2. Goals

- Create a single, consistent import pipeline for YouTube and Spotify.
- Normalize metadata fields across both sources.
- Cache artwork for fast loading and offline support.
- Replace legacy Spotify import logic.
- Provide real-time progress feedback during import.

3. Unified Import Schema

All imported records must conform to the following normalized schema:

- **artist_name** - string
- **song_name** - string
- **album_name** - string
- **artwork_url** - string (cached locally)
- **played_at** - ISO-8601 timestamp
- **track_id** - UUID
- **source** - "spotify" or "youtube"

4. Backend Changes

4.1 New Endpoint

Create a new endpoint:

```
POST /import/unified
```

This endpoint accepts both YouTube and Spotify JSON payloads and performs:

- Schema validation
- Metadata normalization

- Artwork caching
- Deduplication
- Progress event emission

4.2 Artwork Caching

Implement a lightweight `artwork_cache` table:

- **track_id** (PK)
- **artwork_url**
- **cached_at** timestamp

Artwork URLs are stored for fast retrieval and offline support.

4.3 Deduplication

Deduplicate based on:

- `track_id`
- `played_at`

4.4 Progress Tracking

Emit progress events using Server-Sent Events (SSE) or WebSockets. Stages include:

- File validation
- Record normalization
- Artwork caching
- Database insertion
- Completion

5. Ingestion Worker Adjustments

- Detect source type automatically.
- Query external APIs for missing artwork.
- Cache artwork URLs.
- Emit worker-side progress updates.

6. Frontend Updates

6.1 Import UI

Enhance the import UI to:

- Accept both YouTube and Spotify files.

- Display a progress bar.
- Show dynamic status messages.
- Display errors for failed records.

6.2 Artwork Display

Use cached artwork immediately after import.

7. Testing Strategy

- Unit tests for schema validation.
- Integration tests for artwork caching.
- Regression tests for legacy Spotify imports.
- UI tests for progress tracking.

8. Deployment Steps

1. Implement unified endpoint.
2. Add artwork caching.
3. Update ingestion worker.
4. Update frontend import UI.
5. Deploy backend and frontend.
6. Run full regression suite.

9. Commit Messages

- feat(import): add unified import endpoint
- feat(import): implement artwork caching
- feat(import): add progress tracking to import flow
- fix(spotify): normalize schema for unified import
- chore(worker): update ingestion worker for unified import

10. Summary

This unified import flow improves reliability, consistency, and user experience by merging YouTube and Spotify imports into a single pipeline, adding artwork caching, and providing real-time progress feedback.